Aaron Dutta

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Education

University of Waterloo Sep 2026 - Apr 2031

BMath, Honours Mathematics with Co-Op *Waterloo, ON*

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL
Libraries/Frameworks: React.js, Next.js, Node.js, FastAPI, RestAPIs, Flask, Hono, Prisma
AI/ML & Data: OpenCV, MediaPipe, NumPy, Redis, Pandas, ONNX Runtime, Pydantic
Tools & Platforms: Git, Docker, PostgreSQL, Azure, Cloudflare, GCP, Vercel, Supabase, Git

Experience

Magic Hour AI (YC W24) Sep 2026 – Present

Software Engineering Intern *San Francisco, CA*

- Designing and shipping API-integrated products that showcase Magic Hour's video-generation capabilities and grow adoption of the public developer API
- Building full-stack demos and tooling with **Hono**, **PostgreSQL**, **Redis**, and **Cloudflare R2/S3**, using **ffmpeg** for transcoding and media assembly
- Accelerating AI-backed features with **Typesafe AI Jev** and **Meta Muse** while keeping production-oriented data and storage patterns

UW Cube Aug 2026 – Present

Executive Design Engineer *Waterloo, ON*

- Building and maintaining the Cube public website in **Next.js** and **TypeScript** and integrating an AI chatbot by connecting **Python/LangChain** backend services
- Leading **GitHub** collaboration as an executive design engineer, reviewing PRs with software engineers and aligning design and implementation within a cross-functional team
- Designing internal tools to streamline developer workflows, reducing manual steps during building and shipping

Projects

TechniqueTitan | *Python, MediaPipe, OpenCV, NumPy, FastAPI, Pandas, Matplotlib* | Jul 2026 - Present

- Built an AI piano posture coach that turns any device camera into on-demand technique feedback so students can correct hand form without a teacher
- Engineered a **MediaPipe + OpenCV + NumPy** pipeline to extract **21 landmarks** per hand, normalize for camera distance, and score **5 posture criteria**
- Shipped a **FastAPI + React/TypeScript** app with on-device browser MediaPipe and YAML-tunable thresholds, cutting review from a full lesson cycle to seconds
- Built a batch CLI that walks image folders and exports CSV/JSON metrics along with a **45-test pytest** suite across geometry, scoring, and API modules, automating dataset-scale analysis that was previously manual

SkyEye | *Python, Flask, YOLOv8, ONNX Runtime, React, TypeScript* | Aug 2026

- Won RescueHacks 2026 for an SAR triage tool that turns a free-text missing-person report into a Lost Person Behavior search ring, scoring map pins in **5–8s** within a **30s** target
- Developed a **sliding-window YOLOv8n + ONNX Runtime** pipeline to tile images into overlapping windows
- Shipped a **Flask + React/TypeScript** operator console with Gemini Flash (Groq fallback) for typed extraction, and **GCP** for geocoding locations and drawing search zones

Awards

- Winner @ RescueHacks, 160+ hackers Sep 2026
- 2nd Place Overall @ Katy Youth Hacks, 220+ hackers Aug 2026
- 3rd Place Security/Privacy Hack @ STEMist Hacks, 330+ hackers Aug 2026
- 1st Place @ STEMMINDS Hackathon, 60+ hackers Mar 2026